

Project 2: Clinical Feedback Sentiment + Adverse-Event Classification

Goal

Classify short patient feedback/drug reviews by **sentiment** (pos/neg/neutral) and detect **adverse-event mentions** (binary multilabel). Compare:

- **TF-IDF + Linear SVM/LogReg**
vs.
- **Transformer fine-tuning** (DistilBERT; optionally **BioClinicalBERT**).

Data

- Use **Drugs.com Reviews** (Kaggle) or create a **balanced 10k subset** (or synthesize ~8–12k snippets combining symptom lexicons and polarity cues).
<https://www.kaggle.com/datasets/mohamedabdelwahabali/drugreview/data>
- Provide a small AE lexicon (e.g., “nausea”, “headache”, “rash”, “dizziness”, “palpitations”) to bootstrap weak labels.

How to prepare

1. Download dataset (or generate synthetic 10k CSV: text, sentiment, adverse_event).
2. Clean (lowercase, basic de-dup, length filters).
3. Split 80/10/10.

Starter code (classical baseline + transformer)

Provided separately

Extensions

- **spaCy** NER to detect drug, symptom entities → features.
- **Adverse event detection** as second head/classifier (binary) or multi-label.

Evaluation rubric (100 pts)

- Translate business problem to an optimal technical solution, including approach, options explored, logical sequencing of the workflow and tasks involved: **20**
- Data prep & label balance/quality: **10**
- Baseline vs. transformer comparison (macro-F1, confusion matrices): **20**
- Entity/lexicon features & error analysis: **10**
- Exploration of **healthcare-specific models** (BioClinicalBERT etc.) with discussion: **10**
- Presentation clarity: **10**
- Teamwork: **10**
- Timely completion and submission: **10**

References

- <https://www.kaggle.com/code/habibulbasher01644/drug-review-dataset-sentiment-analysis>